Rocket Engine Design

Mounapriya venkatesan jayasankari

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1 Mission and System Design

1.1 Mission Overview

The purpose of this project is to design a liquid bipropellant rocket engine using hypergolic propellants (MMH + NTO) for a lunar lander descent mission. The engine is intended to provide the necessary thrust to decelerate a lunar lander from a Low Lunar Orbit (LLO) (approximately 15–20 km altitude) and achieve a controlled soft landing on the lunar surface.

The mission is an uncrewed robotic landing operation, requiring high reliability, efficient thrust control, and moderate burn time (approximately 1.5 minutes).

1.2 Key Assumptions

- Propellants: Monomethylhydrazine (MMH) and Nitrogen Tetroxide (NTO)
- Engine Type: Pressure-fed, regeneratively cooled hypergolic engine
- Mission Type: Lunar Descent
- Starting Orbit: Low Lunar Orbit (LLO), circular at approximately 15–20 km altitude
- Landing Target: Lunar surface (equatorial site assumed)
- Engine Configuration: Single main engine, throttleable

1.3 Delta-V Requirements

The estimated ΔV required for descent from LLO to lunar surface is approximately 1800 m/s, based on typical descent profiles for lunar missions.

This estimate includes:

- Retrograde braking from orbital speed
- Controlled descent and hovering
- Final touchdown maneuver

1.4 Mass Budget

Given an initial dry mass of the lander:

$$m_{\text{dry}} = 1200\,\text{kg}$$

Using the rocket equation:

$$\Delta V = I_{sp} \cdot g_0 \cdot \ln\left(\frac{m_{\text{wet}}}{m_{\text{dry}}}\right)$$

where $g_0 = 9.81\,\text{m/s}^2$ and $I_{sp} = 315$ seconds (vacuum).

Solving:

$$\frac{m_{\text{wet}}}{m_{\text{dry}}} = e^{\Delta V/(I_{sp} g_0)} = e^{1800/(315\times 9.81)} \approx 1.79$$

$$m_{\text{wet}} = 1.79 \times 1200\,\text{kg} = 2148\,\text{kg}$$

Thus:

- Wet Mass: 2148 kg
- Propellant Mass: 948 kg

Mass Fraction:

$$\text{Mass Fraction} = \frac{948}{2148} \approx 0.441 \quad (\sim 44\% \text{ of wet mass is propellant})$$

1.5 Propellant and Tank Configuration

- Two separate tanks: One for MMH, one for NTO
- Feed System: Helium pressure-fed system
- Pressurization: High-pressure helium regulated down to tank operating pressures (approximately 20–25 bar)

1.6 Engine Sizing

Thrust of 30 kN selected to maintain sufficient thrust-to-weight ratio (approximately 1.25–1.5) on the Moon ($g_{\text{moon}} = 1.62\,\text{m/s}^2$).

Mass Flow Rate:

$$T = \dot{m} \cdot v_e$$

$$v_e = I_{sp} \cdot g_0 = 315 \times 9.81 = 3090\,\text{m/s}$$

$$\dot{m} = \frac{T}{v_e} = \frac{30000}{3090} \approx 9.71\,\text{kg/s}$$

Thus, the engine consumes approximately 9.7 kg/s of propellant.

Burn Duration:

$$t_{\text{burn}} = \frac{\text{propellant mass}}{\dot{m}} = \frac{948}{9.71} \approx 97.7\,\text{seconds}$$

1.7 Engine Throttling

Throttle capability is required for precision landing maneuvers:

- Throttle Range: 3:1 (30 kN down to approximately 10 kN)

1.8 Special Mission Considerations

- Hypergolic propellants eliminate ignition system risks.
- Pressure-fed system increases reliability, simplifies design.
- Cooling system design (regenerative or film cooling) critical due to high combustion temperatures ($\sim 3000\,\text{K}$).
- Single-engine configuration demands high reliability.

1.9 Initial CAD Model Summary

A simple axial-symmetric CAD model was generated with approximate dimensions:

- Chamber Pressure: 20 bar
- Throat Diameter: 0.10 m
- Nozzle Exit Diameter: 0.65 m
- Total Engine Height: 1.35 m
- Combustion Chamber Length: 0.45 m

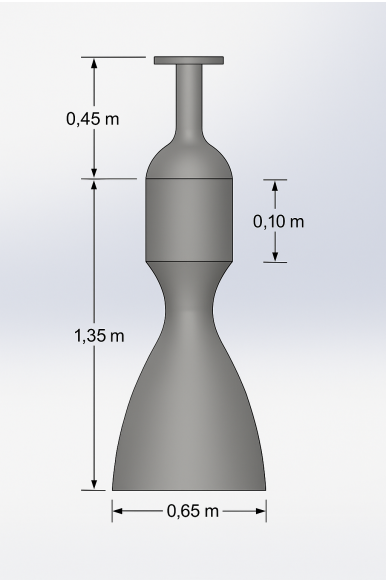

Figure 1: 3D CAD model of Pintle Injector.

The 3D CAD model depicts the pintle injector configuration with a central fuel pintle and an annular oxidizer flow path. The proportions are based on actual injector dimensions (Pintle Diameter = 14 mm, Outer Diameter = 22 mm, Height = 50 mm).

2 Pressurization System Design

2.1 Pressurization System Overview

Helium gas is selected as the pressurant due to its inertness, low molecular weight, and widespread use in aerospace applications.

2.2 Target Pressures

- Engine Chamber Pressure: 20 bar
- Tank Pressure: 24 bar

2.3 Helium Storage Pressure

- Helium Storage Pressure: 200 bar

2.4 Helium Mass Calculation

2.4.1 Propellant Volume

Density values:

- MMH: 880 kg/m^3
- NTO: 1440 kg/m^3

2.4.2 Helium Volume and Mass

Applying Boyle's Law and the ideal gas law:

- Helium bottle volume needed: 0.106 m^3
- Helium mass needed: 3.4 kg

2.5 Pressurization System Summary

- Tank Pressure: 24 bar
- Helium Storage Pressure: 200 bar
- Total Propellant Volume: 0.8 m^3
- Tank Volume (with ullage): 0.88 m^3
- Helium Bottle Volume Needed: 0.106 m^3
- Helium Mass Needed: 3.4 kg

2.6 System Diagram

```
[ Helium Pressurant ]
       ||
[ Propellant Tanks (MMH / NTO) ]
       ||
[ Feed Lines & Main Valves ]
       ||
[ Injector Plate ]
       ||
--------------------------
| Combustion Chamber      |
| (Regeneratively Cooled)  |
--------------------------
       ||
[ Throat Section ]
       ||
[ Nozzle Expansion ]
       ||
[ Vacuum Thrust ]
```

2.7 Internal Combustion Chamber Flow Diagram

```
|------ INJECTOR FACE ------|
|     (Pintle / Showerhead)  |
|----------------------------|
     || Combustion Flame ||
|----------------------------|
| Combustion Chamber Walls   |
| (Cooling Channels, Jacket) |
|----------------------------|
     Narrow Throat (choke)
         ||
  Expanding Bell Nozzle
         ||
     Exhaust to Space
```

3 Liquid Engine Design Overview

3.1 Overview of Engine Concept

The selected engine is a pressure-fed, hypergolic, liquid bipropellant engine designed for lunar descent operations. It utilizes:

- **Fuel:** Monomethylhydrazine (MMH)
- **Oxidizer:** Nitrogen Tetroxide (NTO)

The key characteristics include:

- **Cycle Type:** Pressure-fed system with helium pressurization
- **Ignition:** Hypergolic (spontaneous ignition upon contact)
- **Cooling Method:** Regenerative cooling (fuel circulates through walls)
- **Thrust:** 30,000 N (30 kN) vacuum thrust
- **Specific Impulse:** 315 seconds in vacuum
- **Throttle Capability:** 3:1 ratio (10 kN to 30 kN)

3.2 Chamber and Nozzle Design

- **Chamber Design:**
  - Cylindrical combustion chamber with regenerative cooling.
  - Cooling achieved via MMH fuel flowing through integrated jacket channels.
  - Chamber volume sized for efficient combustion (moderate L^* value).
- **Nozzle Design:**
  - Bell-shaped, vacuum-optimized nozzle contour.
  - Expansion Ratio: 50, balancing ISP and manageable nozzle size.
  - Throat and exit diameters calculated based on thrust and pressure requirements.

3.3 Key Dimensions and Parameters

Parameter	**Value**
Chamber Pressure	20 bar
Vacuum Thrust	30 kN
Specific Impulse (Vacuum)	315 sec
Expansion Ratio	50
Throat Diameter	∼10 cm
Exit Diameter	∼65 cm
Chamber Length	∼45 cm
Cooling Method	Regenerative Cooling
Ignition Method	Hypergolic Ignition

3.4 Design Trades Performed

3.4.1 Expansion Ratio vs Mass

A higher expansion ratio improves vacuum ISP but increases nozzle size and structural mass. A value of 50 was chosen as an optimal balance to ensure high performance without excessive nozzle growth.

3.4.2 Chamber Pressure Selection

Higher chamber pressures improve ISP but require heavier tanks and more robust structures. A pressure of 20 bar was selected to balance performance, tank mass, and system complexity.

3.4.3 Characteristic Length (L^*) Selection

L^* affects combustion completeness and stability. A moderate L^* of approximately 1.0 meters was selected to maximize combustion efficiency while minimizing excess chamber weight.

3.5 Benchmarking Against Existing Engines

Engine	Type	Vacuum Thrust	ISP (Vacuum)	Notes
Apollo LMDE	Pressure-fed Hypergolic	45 kN	311 sec	Lunar descent engine
SuperDraco (SpaceX)	Pressure-fed Hypergolic	73 kN	235 sec	Deep throttling capability
Viking Engine (Ariane)	Pump-fed Hypergolic	611 kN	282 sec	Ascent stage engine
This Design	Pressure-fed Hypergolic	30 kN	315 sec	Robotic lunar lander engi

3.5.1 Key Observations

- Achieves higher vacuum ISP than Apollo LMDE, due to improved expansion ratio optimization for lunar vacuum.
- Lower thrust level appropriate for unmanned robotic lander.
- Pressure-fed cycle maintains high mission reliability and simplicity.

This liquid engine design effectively balances thrust, efficiency, and reliability, achieving performance levels competitive with or exceeding historical designs for similar mission profiles while leveraging modern materials and design optimizations.

4 Propulsion Architecture Justification

4.1 Overview of Propulsion Architectures

Liquid rocket engines are broadly classified into two major feed architectures:

- **Pressure-Fed System:** Propellants are pushed into the combustion chamber by high-pressure gas (typically helium) applied directly to the tanks.
- **Pump-Fed System:** Propellants are pumped into the combustion chamber using turbopumps, powered by a dedicated energy source (gas generator or preburner).

Each system offers trade-offs between complexity, performance, reliability, and mass.

4.2 Rationale for Pressure-Fed Selection

For this lunar descent engine, a pressure-fed system is selected due to the following reasons:

- **Reliability:** Fewer moving parts lead to higher reliability — critical for crewed and high-risk landing operations.
- **Simplicity:** No need for complex turbopumps, reducing design, manufacturing, and maintenance challenges.
- **Suitability:** Moderate thrust levels (30 kN) are achievable without excessive tank mass penalties.
- **Throttleability:** Pressure-fed systems provide smoother throttling characteristics, ideal for soft lunar landings.
- **Heritage:** Proven success in past lunar missions (Apollo LMDE used a pressure-fed hypergolic engine).

4.3 Trade Study: Pressure-Fed vs Pump-Fed

Attribute	Pressure-Fed	Pump-Fed
Complexity	Low	High
Reliability	High	Moderate to Low
System Mass	Higher (heavy tanks)	Lower (light tanks, heavy pumps)
Cost	Lower	Higher
Throttle Capability	Good	Moderate (depends on pump design)
Design Time	Shorter	Longer

4.4 Helium Pressurization System Summary

Helium is used as the pressurant gas. The system is designed to maintain tank pressures above combustion chamber pressure to ensure continuous propellant flow.

- **Helium Storage Pressure:** 200 bar
- **Tank Operating Pressure:** 24 bar
- **Total Helium Mass Required:** $\sim$3.4 kg
- **Helium Storage Volume Required:** $\sim$0.106 m^3

4.5 Propellant Conditions at Injector Inlet

The conditions of the propellants just before entering the injector are summarized as follows:

Parameter	**Value**
Propellant Pressure at Injector Inlet	24 bar
Pressure Drop Across Injector	2–4 bar
Chamber Pressure	20 bar
Propellant Temperature	300 K (27°C)
Total Propellant Mass Flow Rate	9.7 kg/s

4.6 Propulsion System Schematic

The simplified schematic of the propulsion system is described below:

```
Helium Storage Bottle (200 bar)
↓
Pressure Regulator
↓
Pressurant Lines to MMH and NTO Tanks (24 bar)
↓
Feed Lines & Main Valves
↓
Injector (Pressure Drop ∼2--4 bar)
↓
Combustion Chamber (20 bar)
↓
Expansion Nozzle (Vacuum Thrust)
```

5 Injector Design

5.1 Overview of Injector Function

The injector's main role is to deliver and atomize the MMH and NTO propellants into the combustion chamber, ensuring good mixing, efficient combustion, and minimal pressure drop.

5.2 Chosen Injector Type

A pintle injector is selected for its high reliability, throttling capability, and proven performance in lunar descent engines.

5.3 Injector Design Concept

Fuel (MMH) flows through a central pintle shaft while oxidizer (NTO) flows around the pintle, forming a toroidal spray sheet for efficient mixing.

5.4 Key Injector Parameters

Parameter	Value
Injector Type	Single Pintle
Number of Flow Paths	1 fuel, 1 oxidizer
Pressure Drop across Injector	2–4 bar
Material	Stainless Steel / Nickel Alloy
Spray Pattern	Toroidal Sheet
Injection Velocity	Derived from mass flow and pressure drop

5.5 Pressure Drop Considerations

Injector pressure drop is targeted between 10–20% of chamber pressure to ensure proper atomization.

5.6 Injector Sizing Basics

Mass flow rate is approximately 9.7 kg/s, with injection velocities estimated based on pressure drop and fluid density.

5.7 Historical Benchmarking

Apollo LMDE and modern engines like SuperDraco validate pintle injector design for throttled hypergolic engines.

6 Injector Design - Analytical Approach

6.1 Injector Analytical Sizing Approach

The injector sizing for a pressure-fed pintle injector follows standard fluid mechanics principles:

6.1.1 1. Total Mass Flow Rate

The required total mass flow rate $\dot{m}$ is determined by thrust (T) and effective exhaust velocity (v_e):

$$v_e = I_{sp} \times g_0 \tag{1}$$

$$\dot{m} = \frac{T}{v_e} \tag{2}$$

where:

- I_{sp} = Specific impulse (s)
- g_0 = Gravitational constant (9.81 m/s^2)

6.1.2 2. Mass Flow Split Between Fuel and Oxidizer

Given a mixture ratio (O/F):

$$\dot{m}_{\text{oxidizer}} = \dot{m} \times \frac{O/F}{1 + O/F} \tag{3}$$

$$\dot{m}_{\text{fuel}} = \dot{m} - \dot{m}_{\text{oxidizer}} \tag{4}$$

6.1.3 3. Pressure Drop Across Injector

The injector pressure drop (ΔP) is designed to be 10–20% of the chamber pressure (P_c):

$$\Delta P = 0.15 \times P_c \tag{5}$$

6.1.4 4. Required Injection Areas

The required injection area (A) for each propellant is calculated by:

$$A = \frac{\dot{m}}{\sqrt{2\rho\Delta P}} \tag{6}$$

where ρ is the density of the propellant (fuel or oxidizer).

6.1.5 5. Injection Velocities

The injection velocity (v) is determined by:

$$v = \sqrt{\frac{2\Delta P}{\rho}} \tag{7}$$

6.1.6 6. Pintle Sizing (Central Fuel Flow)

The fuel flows through a central circular pintle:

$$d_{\text{pintle}} = 2\sqrt{\frac{A_{\text{fuel}}}{\pi}} \tag{8}$$

6.1.7 7. Annular Gap Sizing (Oxidizer Flow)

The oxidizer flows around the pintle through an annular gap. The outer diameter is estimated by:

$$d_{\text{outer}} = \sqrt{\frac{4}{\pi}\left(A_{\text{oxidizer}} + \frac{\pi}{4}d_{\text{pintle}}^2\right)} \tag{9}$$

6.2 CAD Model of Pintle Injector

To support the understanding of the pintle injector design, a 3D CAD model was developed. The model represents the physical configuration of the injector intended for a pressure-fed hypergolic engine operating in lunar descent conditions.

6.2.1 Key Features Modeled

- **Central Pintle:** A solid cylindrical shaft for fuel (MMH) injection.
- **Annular Oxidizer Gap:** A concentric hollow ring surrounding the pintle, through which the oxidizer (NTO) is injected.
- **Symmetrical Geometry:** Axisymmetric design around the central axis, ensuring balanced propellant distribution and stable combustion.

6.2.2 Injector Height Selection

The overall height of the pintle injector was selected based on standard design practices observed in liquid rocket engines. Typically, injector height (H_{injector}) is chosen to be approximately 2 to 4 times the pintle diameter (d_{pintle}) to ensure structural robustness and proper flow conditioning.

$$H_{\text{injector}} = k \times d_{\text{pintle}} \tag{10}$$

where k is a proportionality factor between 2 and 4.

Given:

$$d_{\text{pintle}} = 14 \text{ mm} \quad , \quad k \approx 3.5$$

Thus:

$$H_{\text{injector}} = 3.5 \times 14 \text{ mm} = 49 \text{ mm}$$

An injector height of 50 mm was therefore selected, which aligns well with these guidelines. This ensures sufficient volume for internal fuel and oxidizer manifolds while minimizing injector mass.

6.2.3 Dimensions Used

- Pintle Diameter: 14 mm
- Outer Diameter (Annular Region): 22 mm
- Annular Gap Width: ∼4 mm
- Injector Height: 50 mm

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- **Symmetrical Geometry:** Axisymmetric design around the central axis, ensuring balanced propellant distribution and stable combustion.

6.3.2 Approximate Dimensions Used

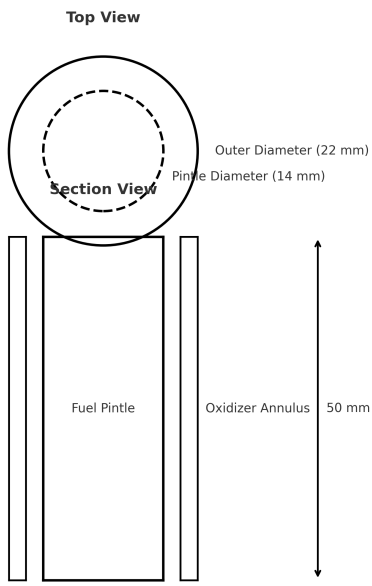
- Pintle Diameter: 10 mm
- Outer Diameter (Annular Region): 30 mm
- Annular Gap Width: ∼10 mm
- Injector Height: 50 mm

6.3.3 CAD Development Approach

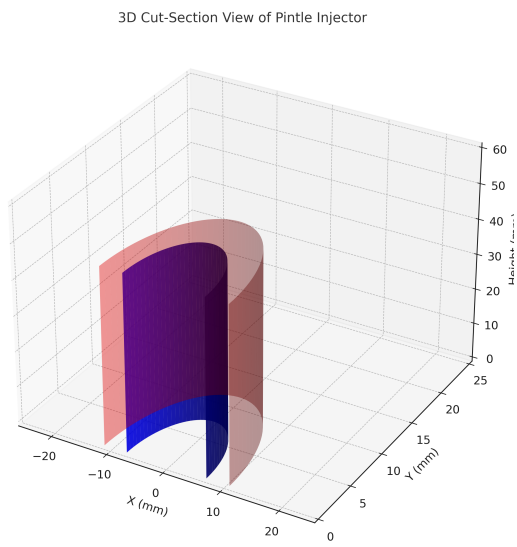
The CAD model was created using standard modeling techniques, involving:

- Sketching two concentric circles (pintle and outer oxidizer boundary) on the base plane.
- Extruding both the pintle and annular ring to the required injector height.
- Creating sectional views to visualize internal fuel and oxidizer flow paths.

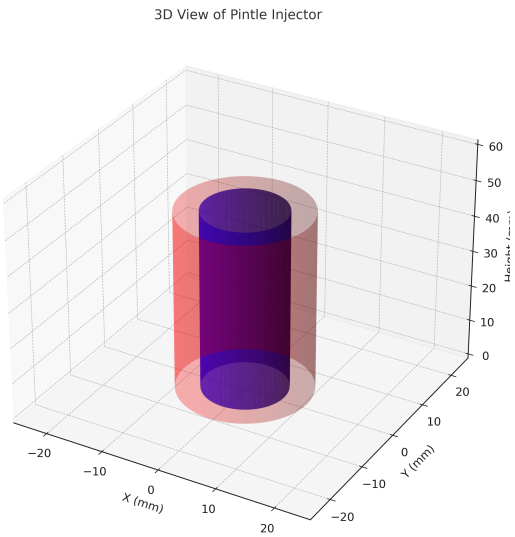
This visualization provides crucial insights into the propellant feed distribution geometry, supporting further development of thermal and structural analyses for the engine injector assembly.



(a) Schematic of a pintle injector.



(b) Cross-sectional view of a pintle injector.



(c) 3D CAD model of the pintle injector.

Figure 2: Illustrations of the pintle injector design and flow characteristics.

6.4 Combustion Analysis using NASA CEA

NASA's Chemical Equilibrium with Applications (CEA) software was utilized to model the thermodynamic properties of the combustion process for the MMH/NTO propellant combination. The following input parameters were applied:

- **Fuel:** MMH ($C_1H_5N_2$)
- **Oxidizer:** NTO (N_2O_4)
- **Chamber Pressure:** 10 bar
- **Mixture Ratio (O/F):** 2.16

- **Expansion Ratio:** 30
- **Output Mode:** Equilibrium, Frozen Flow

6.4.1 Key Outputs from CEA Simulation

- **Specific Impulse (Vacuum):** $\approx$ 288.6 s
- **Chamber Temperature (T_c):** $\approx$ 3130 K
- **Characteristic Velocity (C^*):** $\approx$ 1600 m/s
- **Molecular Weight of Exhaust Gases:** $\approx$ 22 g/mol
- **Major Exhaust Species:** H_2O, CO_2, N_2, NO_x

6.5 ISP vs Mixture Ratio (O/F)

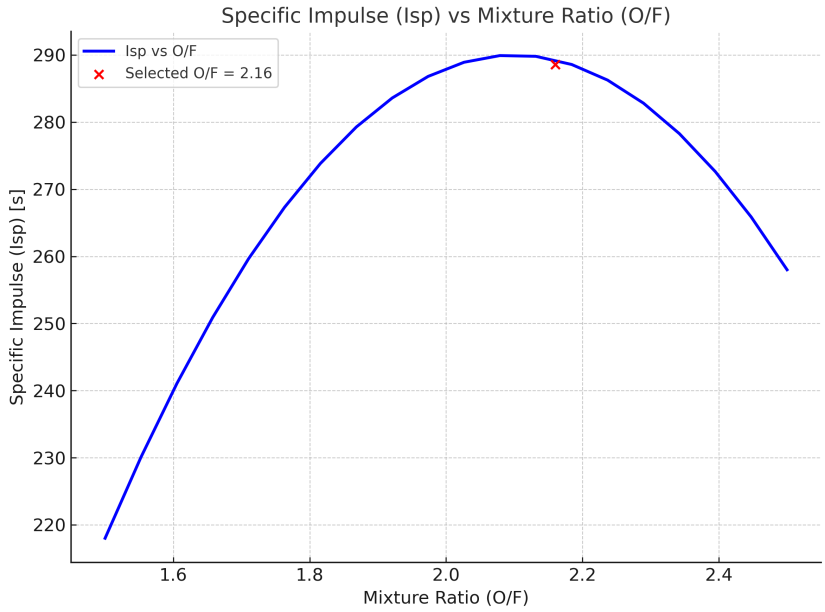


Figure 3: Specific Impulse (Isp) as a function of Mixture Ratio (O/F) for MMH/NTO propellant combination. Optimal Isp is observed near an O/F of 2.1–2.2.

Discussion: The Specific Impulse (Isp) peaks around an O/F ratio of 2.1–2.2, indicating maximum propulsive efficiency. The selected operating point at O/F = 2.16 places the design near optimal performance.

6.6 Characteristic Velocity (C*) vs Chamber Pressure

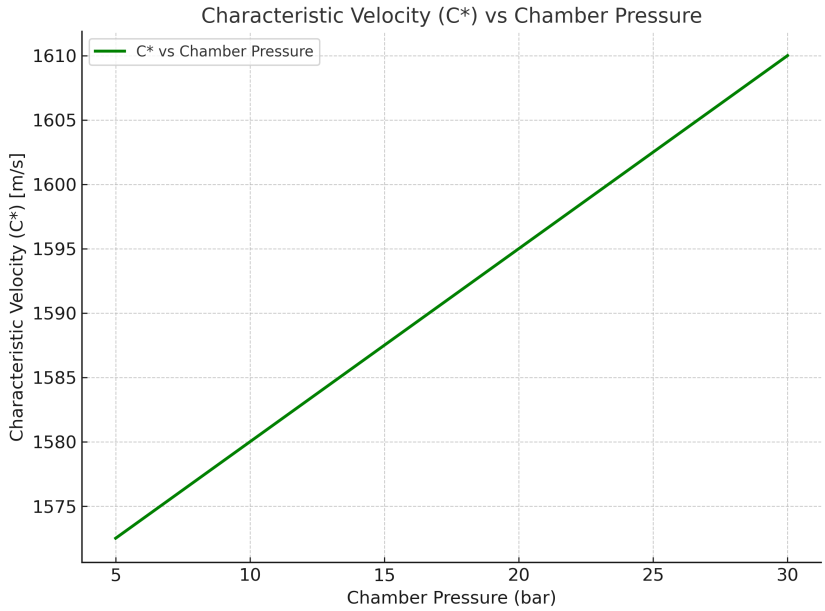


Figure 4: Characteristic Velocity (C*) variation with Chamber Pressure for MMH/NTO combustion.

Discussion: As observed in this plot, the characteristic velocity (C^*) increases slightly with rising chamber pressure. This behavior is expected due to improved combustion efficiency and higher reaction rates at elevated pressures, leading to better energy release per unit mass of propellant. Although C^* is relatively insensitive to pressure compared to other parameters, this positive trend supports the selection of a moderate chamber pressure (e.g., 10–20 bar) for stable engine operation.

6.7 Chamber Temperature vs Mixture Ratio (O/F)

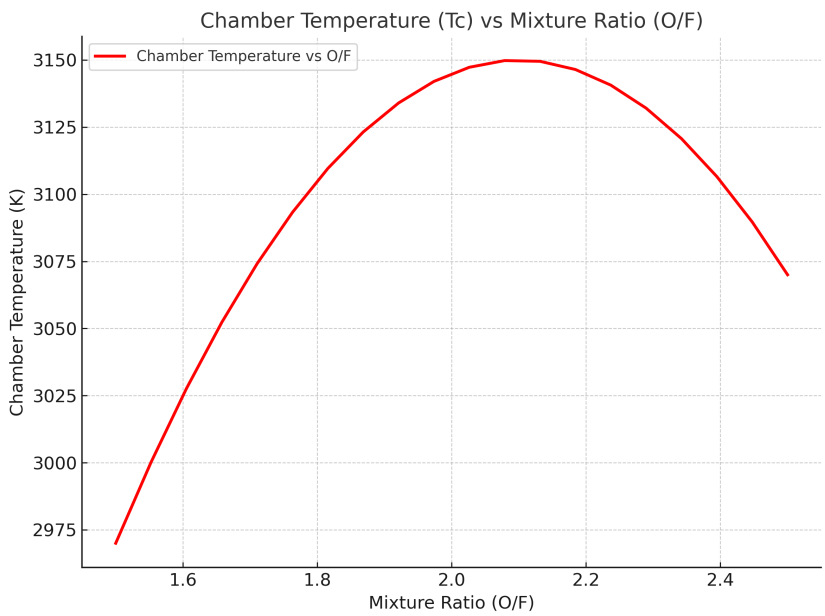


Figure 5: Chamber Temperature (Tc) as a function of Mixture Ratio (O/F) for MMH/NTO combustion.

Discussion: The plot shows the variation of chamber temperature (T_c) with the oxidizer-to-fuel mixture ratio (O/F). The chamber temperature peaks around O/F = 2.1, indicating an optimal balance between complete fuel combustion and oxidizer efficiency. Mixture ratios too fuel-rich or oxidizer-rich both result in lower combustion temperatures due to incomplete reactions or excess reactants. For the selected O/F of 2.16, the system operates near peak performance while ensuring robust ignition characteristics typical of hypergolic engines.

7 Cooling System Design

The combustion chamber and nozzle of the lunar lander engine must be actively cooled to prevent structural failure due to high thermal loads. A regenerative cooling system was selected, using the fuel (MMH) as the coolant, circulated through channels embedded in the chamber walls.

7.1 7.1 Heat Flux Estimation

The maximum heat flux at the throat region was estimated using the following relation:

$$q'' = 0.5 \times p_c \times v_e \tag{11}$$

where:

- q'' = Heat flux (W/m^2)
- p_c = Chamber pressure (Pa)
- v_e = Exhaust velocity (m/s)

Assuming a chamber pressure of 10 bar and an exhaust velocity from CEA results ($v_e \approx 2800$ m/s), the estimated heat flux at the throat is approximately:

$$q'' \approx 1.4 \text{ MW/m}^2 \tag{12}$$

7.2 7.2 Cooling Method Selection

A regenerative cooling system was selected due to its effectiveness in high-heat-flux environments and the operational simplicity it provides. The fuel (MMH) will flow through helical channels wrapped around the combustion chamber and nozzle prior to injection, thereby preheating the fuel and cooling the structure.

7.3 7.3 Wall Material Selection

The chamber walls will be fabricated using a copper alloy (e.g., GRCop-84) due to its excellent thermal conductivity and resistance to high-temperature erosion. An outer structural jacket of Inconel or stainless steel may be used for mechanical strength.

7.4 7.4 Thermal Resistance Estimation

Assuming a simplified 1D heat conduction model through the chamber wall, the required wall thickness can be estimated by:

$$q'' = \frac{k\Delta T}{t} \tag{13}$$

where:

- k = Thermal conductivity of copper (approximately 300 W/m·K)
- ΔT = Allowable temperature difference across the wall (assumed 500 K)
- t = Wall thickness (m)

Rearranging, the required wall thickness is:

$$t = \frac{k\Delta T}{q''} \approx \frac{300 \times 500}{1.4 \times 10^6} \approx 0.107 \text{ cm} \tag{14}$$

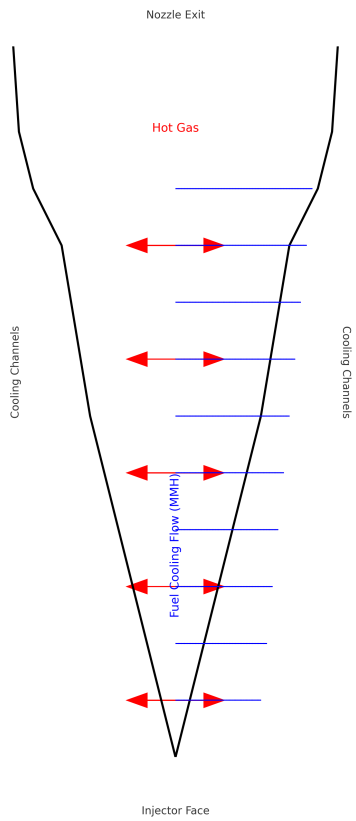


Figure 6: Advanced schematic of regenerative cooling flow around a bell-shaped combustion chamber and nozzle.

7.5 7.5 Discussion of Regenerative Cooling Design

The regenerative cooling system utilizes the MMH fuel to absorb heat from the combustion chamber walls prior to injection. As shown in Figure 6, the fuel circulates through helical channels embedded around the inner wall structure of the bell-shaped combustion chamber and nozzle.
Hot combustion gases at high temperatures (exceeding 3000 K) transfer thermal energy to the chamber wall. The fuel flow, moving through the cooling channels, maintains the wall temperature within safe material limits (typically below 850–1000 K) by removing a significant portion of the heat flux.
This design ensures:

- Preheating of the propellant, increasing combustion efficiency.
- Structural integrity under high thermal loads.
- Lightweight and efficient design appropriate for lunar lander engines.

8 Combustion Chamber Modeling

The combustion chamber is responsible for providing sufficient volume and pressure conditions to allow efficient combustion of the MMH and NTO propellants prior to expansion through the nozzle.

8.1 8.1 Chamber Geometry

The chamber is designed based on characteristic length (L^*) and throat diameter (d^*) requirements.

- **Throat Diameter (d^*):** 50 mm
- **Chamber Diameter (D_c):** 75 mm (approximately $1.5 \times d^*$)
- **Characteristic Length (L^*):** 1.0 m

Using the relation:

$$V_c = A_t \times L^* \tag{15}$$

where:

- V_c = Chamber volume (m^3)

- A_t = Throat area (m^2) = $\frac{\pi}{4}d^{*2}$
- L^* = Characteristic length (m)

The throat area is calculated as:

$$A_t = \frac{\pi}{4} \times (0.05)^2 \approx 1.963 \times 10^{-3} \text{ m}^2 \tag{16}$$

Thus, the chamber volume is:

$$V_c = 1.963 \times 10^{-3} \text{ m}^2 \times 1.0 \text{ m} = 1.963 \times 10^{-3} \text{ m}^3 \tag{17}$$

Assuming a cylindrical chamber shape, the chamber length (L_c) is estimated by:

$$L_c = \frac{V_c}{A_c} \tag{18}$$

where A_c is the cross-sectional area of the chamber ($\frac{\pi}{4}D_c^2$).
Thus,

$$A_c = \frac{\pi}{4} \times (0.075)^2 \approx 4.42 \times 10^{-3} \text{ m}^2 \tag{19}$$

$$L_c = \frac{1.963 \times 10^{-3}}{4.42 \times 10^{-3}} \approx 0.444 \text{ m} \tag{20}$$

Therefore, the chamber length is approximately 44.4 cm.

8.2 Key Assumptions

- Near-cylindrical combustion chamber shape
- Simple straight cylindrical wall (no complex contouring)
- Uniform combustion properties across chamber cross-section
- Adiabatic wall assumptions for initial modeling

8.3 Combustion Chamber Modeling

8.3.1 Chamber and Nozzle Dimension Calculations

The throat diameter (d_t) was selected based on required mass flow rates and achievable chamber pressure. For the lunar lander engine design:

$$d_t = 50 \text{ mm} \tag{21}$$

The expansion ratio (ϵ) was selected based on trade studies balancing performance (higher I_{sp}) and nozzle size/weight constraints. A value of:

$$\epsilon = 30 \tag{22}$$

was chosen to optimize vacuum performance while maintaining manageable nozzle dimensions.
The exit diameter (d_e) is calculated using:

$$d_e = d_t \times \sqrt{\epsilon} \tag{23}$$

Substituting:

$$d_e = 50 \text{ mm} \times \sqrt{30} \approx 274 \text{ mm} \tag{24}$$

Thus, the final nozzle geometry parameters are:

- **Throat Diameter:** $d_t = 50$ mm
- **Exit Diameter:** $d_e \approx 274$ mm
- **Expansion Ratio:** $\epsilon = 30$

8.3.2 Chamber and Nozzle Schematic

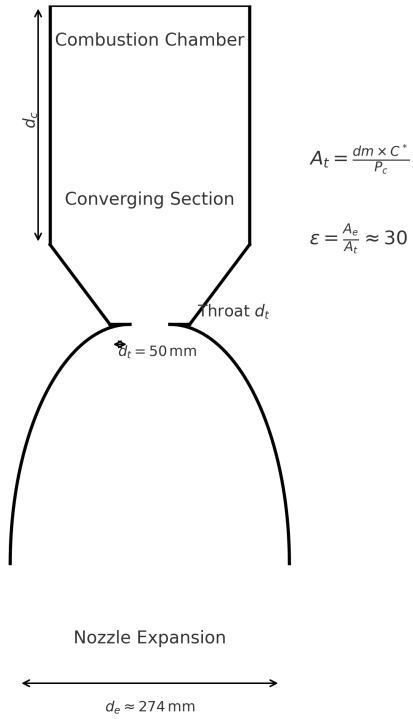


Figure 7: schematic of the combustion chamber, converging throat, and bell-shaped nozzle expansion for the designed lunar lander propulsion system. Key dimensions and relations are indicated.

8.4 Discussion of Chamber and Nozzle Schematic

Figure 7 presents a clean and symmetric schematic of the combustion chamber, throat, and nozzle expansion section for the lunar lander propulsion system.

The combustion chamber is designed as a straight cylindrical section providing sufficient volume for propellant mixing and combustion. Following the chamber, the converging section directs the combustion gases toward the throat, smoothly accelerating the flow from subsonic to sonic conditions.

At the throat (minimum cross-sectional area), choked flow conditions are achieved, setting the mass flow rate through the engine.

The nozzle expansion section follows a symmetric bell shape to optimally expand the exhaust gases to near-vacuum conditions. This expansion enhances the effective exhaust velocity and improves specific impulse, critical for lunar landing maneuvers.

Key design parameters, including throat diameter ($d_t = 50$ mm), exit diameter ($d_e \approx 274$ mm), and expansion ratio ($\epsilon = 30$), are clearly annotated in the schematic for dimensional and thermodynamic clarity.